

AYUSH KASHYAP

Chandigarh, India · +91 9163519554 · [Email](#) · [GitHub](#) · [LinkedIn](#) · [Portfolio](#)

Summary

AI and backend engineer who has shipped production **LLM systems** — cutting **API latency from 1.8s to under 600ms** and improving **RAG retrieval precision by 28%** at a construction-tech startup. Active open source contributor to **Ray** and **Milvus** with **7+ merged PRs**, including feature-level database engine work. Comfortable owning systems spanning **ETL pipelines**, **multi-agent orchestration**, **WASM-compiled engines**, and **reinforcement learning agents**.

Experience

Software Engineer · Constructure AI

Dec 2025 – Apr 2026 · Remote

- Architected **RESTful microservices** for LLM-driven construction software, reducing **p95 API latency from 1.8s to under 600ms** via Redis caching, connection pooling, and async job queues.
- Built scalable **ETL pipelines** processing **500MB+ BIM/IFC datasets** for downstream LLM context generation and structured retrieval.
- Developed **multi-agent orchestration** with **OpenAI SDK**, coordinating **4+ specialized agents** while reducing cross-agent error propagation by **40%**.
- Implemented **hybrid RAG retrieval** combining **pgvector semantic search** with structured SQL filtering, improving retrieval precision by **28%**.

Open Source Contributions

Ray (ray-project/ray)

4+ merged PRs · [PR #63695](#) · [PR #63952](#)

- Fixed **AlgorithmConfig.to_dict()** for RLlib's New API Stack — stale **train_batch_size** values caused incorrect experiment logging across Ray Tune trials; resolved a read-only property conflict preventing safe round-trips with regression tests covering the DreamerV3 edge case.
- Patched a **critical cluster resilience bug** in Ray Core's C++ OOM killer: **TimeBasedWorkerKillingPolicy** killed the **ServeController** under memory pressure, triggering an infinite OOM death-loop; introduced an **is_system_actor** shield across **protobuf** → **C++ raylet** → **Cython** → **Python** with cross-boundary unit tests.

Milvus (milvus-io/milvus)

3+ merged PRs · [PR #51051](#) · [PR #50666](#) · [PR #51636 \(open\)](#)

- Extended the **filter expression engine** to support bitwise AND, OR, and XOR operators for bitmask queries; modified the full pipeline — **protobuf schema**, **ANTLR parser**, **C++ bitset engine**, and **JSON/array/index execution paths (500+ lines across 8 files)**.
- Implemented a new **Decimal scalar field type** for exact fixed-point numerics (e.g. currency data) end-to-end: **protobuf wire format**, **Go parser**, **C++ segcore storage/indexing/query execution**, **filter expression comparisons and arithmetic** — **1000+ line size/XXL PR** currently in review; all CI checks passing.
- Assigned to issue [#51269](#): extending scalar filter expressions to support **nested/chained arithmetic** operations; ongoing.

Projects

GoChess Engine

[Demo](#)

- Built a chess engine in **Go** with **Minimax + Alpha-Beta Pruning** (60% game-tree reduction, depth-5 in under 200ms); cross-compiled to **WebAssembly** with **Web Workers** for **2–3×** throughput; integrated **Zobrist Hashing** and **Transposition Tables** (100K+ cached positions) and **Quiescence Search**.

PyGit — Distributed Version Control Engine

[GitHub](#)

- Re-implemented core **Git internals** (Blob, Tree, Commit) with **content-addressable storage**, **SHA-1 persistence**, and **O(1)** object lookup across 100+ commit repositories; CLI supports **init**, **add**, **commit**, **log** via DAG-based version tracking.

PacmanRL — Deep Reinforcement Learning Agent

[GitHub](#)

- Trained a **DQN agent** with experience replay achieving stable gameplay in **500 episodes**; custom **reward shaping** cut episode length by **30%** and delivered **2×** higher scores over baselines via an **11-dimensional feature pipeline**.

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, C++, C#, Java

AI / ML: LangChain, LangGraph, RAG, pgvector, Pinecone, Milvus, TensorFlow, DQN, OpenCV, Multi-Agent Systems, Prompt Engineering

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, Microservices, Redis, PostgreSQL, ETL Pipelines, Async Processing

Frontend & Web: React, Next.js, React Native, WebAssembly (WASM), Web Workers

DevOps & Infra: Git, Docker, CI/CD, Distributed Systems, Ray, Performance Profiling

Education

Punjab Engineering College (PEC), Chandigarh

B.Tech in Mechanical Engineering

2023 – 2027

CGPA: 8.23 / 10